A READER'S COMPANION · PREPARED FOR THE FALSIFY BUILD

Backtest Overfitting

Why a good backtest proves almost nothing

Companion notes to Bailey, Borwein, López de Prado & Zhu (2014),
Notices of the American Mathematical Society 61(5), 458–471.

This is not the paper. It is a set of notes to read alongside it: the arguments restated in my own words, the two key results derived rather than asserted, the numbers recomputed for your actual grid size, and a map from each result to the gate in the build that enforces it.

GET THE ORIGINAL

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The working-paper version sits at SSRN abstract 2308659, and the DOI is 10.1090/noti1105. Read the original first. Fourteen pages, written for mathematicians rather than traders, and it assumes nothing about finance beyond the definition of a Sharpe ratio.

ONE-LINE SUMMARY

If you search over enough strategy configurations, an outstanding backtest is guaranteed by arithmetic alone — no market edge required. The authors quantify exactly how many configurations it takes and exactly how much history you would need to tell the two apart.

IN THESE NOTES

THE ARGUMENT

1. What the paper actually claims

Start with the definition the authors use, because everything follows from it. A backtest is a historical simulation: you take an algorithmic rule, replay it over a stretch of past prices, and record the profit and loss it would have produced. You then summarise that path with a statistic — usually the Sharpe ratio.

The trouble begins the moment you run the simulation twice. Nobody backtests one configuration. You try a 20/50 moving-average crossover, then 20/100, then 10/50, then a different holding period, then a stop-loss, then the same thing on a second ticker. Each attempt is a draw from a distribution. The one you keep is the maximum of those draws, and the maximum of a sample of noise is not noise-sized. It grows with the sample.

The paper's three results follow from taking that observation seriously:

- Overfitting is cheap. A small number of configurations — tens, not millions — suffices to manufacture an impressive in-sample Sharpe from a series with zero true edge.
- There is a minimum backtest length. Given the number of configurations tried, you can compute the shortest history for which a given in-sample Sharpe carries any information at all. Below that length, the number is uninterpretable.
- Overfitting is worse than useless. When the underlying process mean-reverts, the configuration selected for maximum in-sample Sharpe tends to deliver negative expected out-of-sample performance. Not zero. Negative.

That third result is the one people miss. The intuition is that an overfit rule has latched onto a specific realised wiggle, and under mean reversion the wiggle is followed by its own reversal. You have not merely fitted noise; you have fitted the part of the noise most likely to invert.

> THE PART THAT MAKES IT A MATHEMATICS PAPER
>
> Because almost nobody reports how many configurations they tried, an investor cannot distinguish a genuine edge from a search artefact even in principle. The information needed for the inference has been withheld. The authors' proposed remedy is disclosure of the trial count — which is why your own README will state N, and why your DSR will use it.

SETUP

2. Notation, kept minimal

Let $r_1, \ldots, r_T$ be the per-period (daily) returns of one strategy configuration, with sample mean μ and sample standard deviation σ. Throughout, SR without a subscript means the sample estimate; SR_n with a subscript means the unknown true value of configuration n. The per-observation Sharpe estimate is

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SR = μ / σ (sample mean over sample standard deviation)
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and the annualised figure everybody quotes is $SR \cdot \sqrt{252}$. Keep the two separate in code. Every formula below is in per-observation units, and mixing them is the most common implementation bug in this entire area.

Suppose you evaluate N configurations. Write SR_1, …, SR_N for their estimates and SR_n for the unknown true values. The null hypothesis of interest is not “this strategy has no edge”. It is:

> H_0: every one of the N configurations has true Sharpe zero, and the best observed result is the maximum of N draws of pure estimation noise.

Under that null, and treating the trials as independent, each SR_n is approximately normal with mean zero and standard error $1/\sqrt{T}$. So the quantity you need is the expected maximum of N standard normals, scaled by $1/\sqrt{T}$.

RESULT ONE

3. How large is the best result from pure noise?

The maximum of N independent standard normals has no closed form, but its asymptotic behaviour is classical extreme-value theory: after centring and scaling it converges to a Gumbel distribution. The authors use the standard sharp approximation for its expectation, with $\gamma \approx 0.5772$ the Euler–Mascheroni constant and Φ^{-1} the normal quantile function:

$$E[\max_n z_n] \approx (1-\gamma)\cdot\Phi^{-1}(1 - 1/N) + \gamma\cdot\Phi^{-1}(1 - 1/(N\cdot e))$$

The leading behaviour is $\sqrt{2 \ln N}$, which is the part worth carrying in your head: the best-of-N result grows like the square root of the logarithm of N. Slowly — but it grows without bound, and it grows fastest in exactly the regime a solo researcher operates in, where N moves from 1 to a few hundred.

Multiply by $1/\sqrt{T}$ to get the expected best per-observation Sharpe, then by $\sqrt{252}$ to annualise. Recomputed here for T = 1008 trading days, which is four years of daily data:

N (configurations)	E[max z]	MinBTL, years for SR* = 1.0	Best annualised SR from pure noise, 4y
10	1.575	2.5	0.79
50	2.276	5.2	1.14
100	2.531	6.4	1.27
500	3.053	9.3	1.53
1,000	3.255	10.6	1.63
5,000	3.688	13.6	1.84
100,000	4.391	19.3	2.20

Computed directly from the expression above; Φ^{-1} via scipy.stats.norm.ppf. MinBTL is the number of years of daily data needed before an in-sample annualised Sharpe of 1.0 becomes distinguishable from the best-of-N noise result.

> **READ THE TABLE ONCE MORE**
>
> A 20×20 parameter grid over two strategy families is N = 800. On four years of daily data, pure noise delivers a best annualised Sharpe of about 1.6. If your headline backtest reports 1.4 on that grid, you have found nothing — you have found less than nothing, because you underperformed the noise. And to make an in-sample annualised Sharpe of 1.0 meaningful at N = 800 you would need roughly ten years of history.

RESULT TWO

4. Minimum backtest length

Invert the previous expression. Fix a target: you want an in-sample annualised Sharpe of SR* to be evidence rather than arithmetic. Set the expected best-of-N noise Sharpe equal to SR* and solve for the sample length in years:

$$\mathrm{MinBTL} \approx \left[\,(1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))\,\right]^2 / SR^{*2}$$

The quadratic is the punishing part. Halving your Sharpe target quadruples the history you need. This is why short backtests on large grids are worthless and why the fix is never “add more data” — you cannot manufacture decades of independent history. The fix is to shrink N, or to stop pretending N is 1.

Three honest ways to shrink N

- Pre-register. Decide the parameter values from theory or from the literature before you look at the data. A 12-month momentum lookback chosen because Moskowitz, Ooi and Pedersen used it is N = 1. The same value chosen because it topped your grid is N = grid.
- Cluster the trials. Adjacent configurations are not independent: SMA(20,50) and SMA(21,50) are nearly the same strategy, so counting both as separate trials overstates N. Estimate an effective N from the correlation structure of the trial returns. This cuts N substantially and is defensible — but it must be computed, not assumed.
- Report N truthfully. N is the number of configurations you evaluated, including everything you abandoned. Deleting the failures from your notebook does not delete them from the statistics.

THE CORRECTION

5. From Sharpe to Deflated Sharpe

The companion result, developed in Bailey and López de Prado's 2014 Journal of Portfolio Management paper, converts all of the above into a single probability. Two ingredients.

5.1 The Sharpe ratio has a standard error

SR is an estimator, so it has a sampling distribution. Lo (2002) gives its asymptotic variance, and the correction for non-normal returns matters because strategy returns are skewed and fat-tailed. With γ_3 the skewness and γ_4 the non-excess kurtosis, the Probabilistic Sharpe Ratio is the probability that the true Sharpe exceeds a benchmark SR*:

$$\mathrm{PSR}(SR^*) = \Phi\left[\,(SR - SR^*)\cdot\sqrt{T-1} \;/\; \sqrt{1 - \gamma_3 SR + ((\gamma_4-1)/4)SR^2}\,\right]$$

Sanity check your implementation here, because the kurtosis convention is the classic bug. For normal returns $\gamma_3 = 0$ and $\gamma_4 = 3$, so the denominator collapses to $\sqrt{1 + SR^2/2}$. If yours does not, you have used excess kurtosis where non-excess was wanted, and every number downstream is wrong by a factor that looks plausible.

5.2 Set the benchmark to the noise level

The move that makes it a deflated Sharpe: instead of testing against zero, test against the expected best-of-N result from Section 3. With V the variance of SR across your N trials,

$$SR_0 = \sqrt{V} \cdot [\ (1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))\ ]$$
$$DSR = PSR(SR_0)$$

DSR is the probability that the strategy's true Sharpe is positive after accounting for the fact that you picked it as the winner of a search, and after accounting for the non-normality of its returns. It is the single number the whole literature is pointing at. A strategy with a 2.1 Sharpe and a DSR of 0.4 is a strategy you do not trade.

RESULT THREE

6. Measuring your own overfitting: PBO

DSR corrects a number. The Probability of Backtest Overfitting audits your procedure. It comes from the same authors' follow-up work on combinatorially symmetric cross-validation, and it answers a sharper question: when my selection rule picks a winner in-sample, how often does that winner land in the bottom half out-of-sample?

The construction

Build M, a T×N matrix holding the return series of every configuration you tried. Split the T rows into S contiguous blocks, S even — sixteen works well. For each of the C(S, S/2) ways to choose half the blocks as in-sample:

- find n^* = argmax of the in-sample Sharpe across the N columns;
- compute the out-of-sample Sharpe of every column on the complementary blocks;
- record the rank ω of column n^* among those N out-of-sample Sharpes, and form the relative rank $r = \omega/(N+1)$ and its logit $\lambda = \ln(r/(1-r))$.

PBO is then $P(\lambda \leq 0)$: the fraction of splits in which the in-sample winner came out below median out-of-sample. With S = 16 that is 12,870 splits, which runs in seconds once the matrix M exists.

> HOW TO READ IT
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> PBO ≈ 0.5 — your selection rule is a coin flip. The search has no predictive content whatsoever.
>
> PBO > 0.5 — worse than a coin flip, which is the empirical signature of the mean-reversion result in Section 1. Your winner is systematically the wrong pick.
>
> PBO < 0.2 — the procedure carries real information. Report the number either way; publishing it about your own work is the credibility move.

Note what PBO is not. It does not tell you the strategy works. It tells you whether the method you used to choose it works. Those are different claims and conflating them is how people end up defending an overfit rule with an overfit validation.

APPLICATION

7. Mapping the paper onto the build

Result in the paper	Gate	What the gate does
Best-of-N noise grows like $\sqrt{2 \ln N}$	G6	1,000 coin-flip strategies through the full pipeline; their Sharpe distribution is your empirical null. Your real result must beat that, not a textbook normal.
Minimum backtest length	G9	Compute MinBTL for your actual N before reporting anything. If T < MinBTL, the headline Sharpe is reported as uninterpretable rather than as a result.
Overfit selection → negative OOS	G8	Purged, embargoed walk-forward. If in-sample and out-of-sample Sharpes are anticorrelated across the grid, you have reproduced the paper's central finding on your own data.
Deflated Sharpe Ratio	G6	DSR with N reported honestly. Calibration check: DSR must reject at least 95% of the coin-flip strategies at $\alpha = 0.05$.
PBO via CSCV	G9	Ship only if PBO < 0.5, and print the number in the README regardless of what it says.
Trial count must be disclosed	–	A trials ledger written by the harness, not by you. Every configuration evaluated gets a row. N is read from the ledger.

A twenty-minute exercise before you write engine code

This reproduces the paper's core claim from scratch and takes almost no code. Do it first; it will change how you read every backtest for the rest of your life.

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1. Generate a pure random walk. No drift, no edge, T = 1008 daily bars.
2. Sweep a moving-average crossover over a 20×20 grid of window pairs.
3. Record the annualised Sharpe of all ~400 surviving configurations.
4. Take the maximum. Plot the full distribution with the max marked.
5. Repeat the whole thing over 200 fresh random walks.

Expected: the best-of-grid Sharpe lands near 1.3–1.6 on almost every path,
and the distribution of maxima is tight. The edge is zero by construction.
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Then run the identical sweep on real SPY data over the same window. If the best real Sharpe sits inside the distribution of maxima you just generated from noise, you have learned something true and slightly bleak about moving-average crossovers.

NEXT

8. Where to go after this paper

Bailey & López de Prado (2014)	The Deflated Sharpe Ratio, Journal of Portfolio Management 40(5), 94–107. The formulas in Section 5, in full.
Bailey, Borwein, López de Prado & Zhu (2016)	The Probability of Backtest Overfitting, Journal of Computational Finance. Where CSCV is defined properly.

López de Prado (2018)	Advances in Financial Machine Learning, chapters 7, 11, 12, 14. Purging, embargo, and the practical apparatus.
Harvey, Liu & Zhu (2016)	...and the Cross-Section of Expected Returns, Review of Financial Studies 29(1). The same multiple-testing problem applied to the published asset-pricing literature. Their conclusion: the honest bar is t > 3.0.
Lo (2002)	The Statistics of Sharpe Ratios, Financial Analysts Journal 58(4). Where the standard error in Section 5.1 comes from.
White (2000)	A Reality Check for Data Snooping, Econometrica 68(5). The bootstrap-based test for whether the best of N beats a benchmark.

These notes paraphrase and reorganise the cited work; they reproduce none of it. The published article is free at the AMS link on page 1 and should be read in the original. Numerical tables here were recomputed independently.